

# $\sigma$ -ADDITIVE PROBABILITY ON ORTHOMODULAR LATTICES: A SURVEY AND AN OPEN PROBLEM

PATRICK S. MAHON

**ABSTRACT.** On a Boolean algebra, a finitely coherent probability extends to a countably additive measure by classical machinery (Carathéodory; Loomis–Sikorski; Stone). On a non-distributive orthomodular lattice (OML) this machinery breaks. We survey the problem and isolate a precise open question, splitting it into two axes the Boolean case fuses: the *extension* axis (does a state extend to a charge on the dual?), a Horn–Tarski feasibility question settled negatively for  $L(H)$ ; and the *descent* axis (does the charge concentrate on the physical points?). The combinatorial route fails —  $\prod_n \text{MO}_2$  is concrete,  $\sigma$ -complete and non-Boolean yet *segregated* (non-distributivity confined to finite blocks) — because a faithful set representation forces the OML Boolean (Theorem 5.2). Reading “no hidden realisation” as not being a mixture of dispersion-free states makes a relational measure a *contextual* state (Lemma 5.3). The open question: *does a non-Boolean,  $\sigma$ -complete, concrete OML carry a  $\sigma$ -additive contextual state whose contextuality is  $\sigma$ -essential* — witnessed by no finite sub-OML? The finite case is inhabited (Wright’s pentagon), so only the  $\sigma$ -essential refinement remains, a compactness failure of CE type. Recent work on Dynkin-systems (Derr–Williamson [9]) settles it *negatively under a Polish/Borel-regularity hypothesis*, leaving the residue — whether a witness can be non-Polish-representable — a descriptive-set-theory question (Remark 5.1).

## 1. INTRODUCTION

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \emptyset \quad \implies \quad \mu(A_n) \rightarrow 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [28] adopted  $\sigma$ -additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is

---

*Date:* June 2026.

Boolean; gluing contexts along shared observations (a categorical colimit) returns:

$$\begin{aligned} \text{contexts mutually compatible} &\Rightarrow \text{Boolean algebra,} \\ \text{incompatible} &\Rightarrow \text{non-distributive OML.} \end{aligned}$$

(Gunji et al. [19]). The OML is *forced* by the context structure, not postulated; for non-quantum examples the incompatibility must be operationally imposed, since classical observables on a common space jointly distribute (Kolmogorov) and stay Boolean. The closed subspaces of a Hilbert space  $H$  form the prototype  $L(H)$ , not the premise.

On a non-distributive OML, Carathéodory's outer measure does not apply. For  $L(H)$  it is rescued by Gleason's theorem [16], with no general-OML analogue known. Non-distributivity *per se* does not remove the two-valued homomorphisms the Boolean engine needs (MO<sub>3</sub> keeps them), but it removes the *guarantee*; for  $L(H)$ ,  $\dim \geq 3$ , they fail outright (Kochen–Specker [27]). Whether enough survive is a question of *concreteness*, not non-distributivity: an OML is *concrete* (Definition 2.7) when it carries an order-determining set of two-valued states (Gudder [18]), and this is the qualifier on which the open problem turns.

The organising split. The question splits into two axes the Boolean case fuses: *extension* (does a state extend to a charge on  $S_0(A)$ ? — settled, none does on  $L(H)$ ) and *descent* (does the charge concentrate on the physical points? — open).

The relational standpoint. Following De Finetti [8] (probability without a presupposed sample space) and Birkhoff–von Neumann (propositions as a lattice, not subsets of a phase space), we start from the propositions and ask what their order alone determines — the measure-theoretic counterpart of pointless topology. The dual is then the McDonald–Bimbó space of *filters* (§2.3), and a positive answer would be a  $\sigma$ -additive probability theory on a non-distributive lattice, the non-Boolean analogue of localic measure theory (made precise in §5 as a *contextual* state; cf. [30]).

## 2. PRELIMINARIES

We collect the lattice-theoretic apparatus; all notions are standard.

### 2.1. Orthomodular lattices.

**Definition 2.1** (Orthocomplemented lattice). A *bounded lattice* is a partially ordered set in which every pair  $a, b$  has a join  $a \vee b$  and meet  $a \wedge b$ , with greatest element  $\mathbf{1}$  and least element  $\mathbf{0}$ . An *orthocomplementation* is a map  $a \mapsto a^\perp$  satisfying, for all  $a, b$ :

- (i)  $a \wedge a^\perp = \mathbf{0}$  and  $a \vee a^\perp = \mathbf{1}$  (complement),
- (ii)  $a^{\perp\perp} = a$  (involution),
- (iii)  $a \leq b \Rightarrow b^\perp \leq a^\perp$  (order-reversing).

**Definition 2.2** (Incomparable; meet-zero; orthogonal). For elements  $a, b$  of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*:  $a \not\leq b$  and  $b \not\leq a$  — neither entails the other (an order condition);
- *meet-zero*:  $a \wedge b = \mathbf{0}$  — no common refinement;
- *orthogonal*, written  $a \perp b$ :  $a \leq b^\perp$  (equivalently  $b \leq a^\perp$ ) — one entails the other's negation, *decisively incompatible*.

Each implies the one above it ( $a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$  incomparable, for  $a, b \notin \{\mathbf{0}, \mathbf{1}\}$ ). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only  $\mathbf{0}$  below it, so  $a \wedge b \in \{\mathbf{0}, a\}$ , and  $a \wedge b = a$  would force  $a \leq b$ ; orthogonality remains strictly stronger.

*Remark 2.3* (Meet-zero versus orthogonal). The implication  $a \perp b \Rightarrow a \wedge b = \mathbf{0}$  reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

**Definition 2.4** (Orthomodular lattice). An orthocomplemented lattice  $A$  is *orthomodular* if

$$a \leq b \implies b = a \vee (a^\perp \wedge b). \quad (1)$$

A *Boolean algebra* is the special case where  $\wedge$  distributes over  $\vee$ ; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is  $L(H)$ , the closed subspaces of a Hilbert space with  $a^\perp$  the orthogonal complement.

**Definition 2.5** (Completeness notions). Two countable-completeness conditions on an OML  $A$  are distinguished throughout, the second strictly weaker. Here  $\bigvee_n a_n$  denotes the *join* (least upper bound) of  $\{a_n\}$  in the order of  $A$ , unique when it exists; “ $\exists \bigvee_n a_n \in A$ ” asserts that it does.

- $A$  is  $\sigma$ -complete if

$$\forall \{a_n\}_{n \in \mathbb{N}} \subseteq A : \quad \exists \bigvee_n a_n \in A$$

(equivalently every countable family has a meet, by  $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$ );

- $A$  is  $\sigma$ -orthocomplete if

$$\forall \{a_n\}_{n \in \mathbb{N}} \subseteq A \ (a_i \perp a_j \text{ for } i \neq j) : \quad \exists \bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so  $\sigma$ -completeness quantifies over a strictly larger domain and is the stronger hypothesis:  $\sigma$ -completeness  $\Rightarrow \sigma$ -orthocompleteness. Only the weaker condition is needed below. For a state  $s$ ,  $\sigma$ -additivity is the requirement that

$$s\left(\bigvee_n a_n\right) = \sum_n s(a_n)$$

for every countable pairwise-orthogonal family  $\{a_n\}$ . Its left side invokes  $\bigvee_n a_n$  *only* for orthogonal families — so  $\sigma$ -orthocompleteness is exactly the hypothesis that makes it well-posed. Full  $\sigma$ -completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

**Example 2.6** ( $\text{MO}_n$ , and the gap made concrete).  $\text{MO}_n$  is the lattice with  $\mathbf{0}$ ,  $\mathbf{1}$ , and  $n$  complementary pairs of incomparable atoms, all sharing only  $\mathbf{0}$  and  $\mathbf{1}$ . The geometric model is  $n$  lines through the origin: each line  $a$  is orthogonal to its perpendicular  $a^\perp$ , while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ( $n \geq 2$ ): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad (\text{meet only at the origin}) \quad \text{yet} \quad a \not\perp b,$$

so meet-zero  $\neq$  orthogonal already here.  $\text{MO}_3$  is the smallest non-distributive OML.

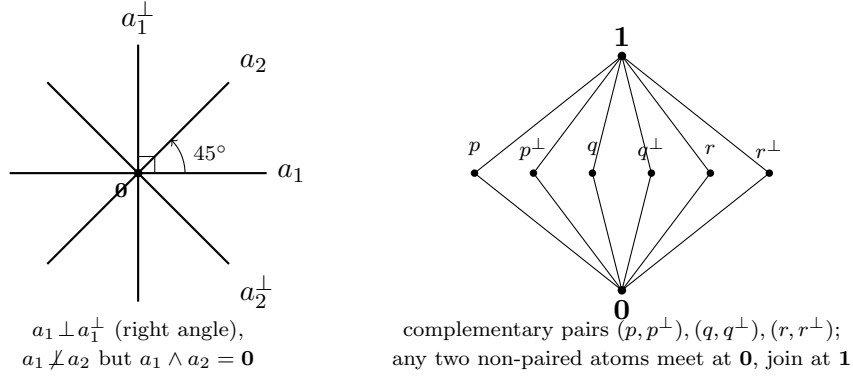


FIGURE 1.  $\text{MO}_n$  illustrated. *Left*: the geometric model for  $\text{MO}_2$  — four atoms as lines through the origin in  $\mathbb{R}^2$ . The pair  $a_1, a_1^\perp$  meets at a right angle (orthogonal), whereas  $a_1, a_2$  meet at  $45^\circ$ : their lattice meet is still  $\mathbf{0}$  (only the origin in common), yet they are *not* orthogonal — the meet-zero  $\neq$  orthogonal gap. *Right*: the Hasse diagram of  $\text{MO}_3$ , the smallest non-distributive OML: three complementary pairs of incomparable atoms between  $\mathbf{0}$  and  $\mathbf{1}$ .

**Definition 2.7** (OMP; order-determining states; concrete logic). An *orthomodular poset* (OMP) weakens Definition 2.4: joins  $a \vee b$  are required only for *orthogonal* pairs. A set  $S$  of states on  $A$  is *order-determining* if it separates the order, i.e.

$$(s(a) \leq s(b) \text{ for all } s \in S) \implies a \leq b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into  $(\mathcal{P}(X), \subseteq, X \setminus (-))$  for some set  $X$ , preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets of outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [18]), and it is the property that separates the open frontier from the settled cases (§5).

*Remark 2.8* (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet  $a \wedge b$  is the greatest  $L$ -member inside  $A \cap B$ , so

$$\underbrace{A \cap B = \emptyset}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \quad \text{but not conversely.}$$

Concreteness does not force the converse:  $\text{MO}_3$  is itself concrete (Definition 2.7; Gudder [18]), so the paradigmatic meet-zero  $\neq$  orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \Rightarrow a \perp b,$$

equivalently: every nonempty  $A \cap B$  contains a nonzero  $L$ -member — the abstract form being Tkadlec’s *Boolean orthoposet* condition [40] (which does *not* mean Boolean *algebra*).

## 2.2. The state/meet-additive/charge ladder.

**Definition 2.9** (State; meet-additive state; charge-extendible). Let  $A$  be an OML. A *state* is a map  $s : A \rightarrow [0, 1]$  with  $s(\mathbf{1}) = 1$  that is additive on *orthogonal* pairs:  $s(a \vee b) = s(a) + s(b)$  whenever  $a \perp b$ ; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on  $\text{MO}_3$ ):

- (1) *State*:  $s(a \vee b) = s(a) + s(b)$  whenever  $a \perp b$ .
- (2) *Meet-additive*:  $s(a \vee b) = s(a) + s(b)$  whenever  $a \wedge b = \mathbf{0}$  (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word “valuation” here: in lattice theory it denotes the modular function  $v(a) + v(b) = v(a \wedge b) + v(a \vee b)$ , a different condition.
- (3) *Charge-extendible*:  $\sum_i s(a_i) \leq 1$  for every pairwise-meet-zero family  $\{a_i\}$  ( $a_i \wedge a_j = \mathbf{0}$ ,  $i \neq j$ ; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

**Example 2.10** (The ladder is strict). On  $\text{MO}_3$  with complementary pairs  $(p, p^\perp), (q, q^\perp), (r, r^\perp)$ , the maximally mixed  $s \equiv \frac{1}{2}$  is meet-additive (clears (2)) yet fails (3) at  $\{p, q, r\}$ :  $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$ . So no state on  $\text{MO}_3$  is charge-extendible.

**Definition 2.11** (Normal and singular states on  $L(H)$ ). On  $A = L(H)$  a state  $s$  is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\left(\bigvee_{i \in I} p_i\right) = \sum_{i \in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

equivalently  $s(\cdot) = \text{tr}(\rho \cdot)$  for a density operator  $\rho$  (Gleason [16],  $\dim \geq 3$ );

- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

### 2.3. The McDonald–Bimbó dual space.

**Definition 2.12** (Dual space; representation map). The McDonald–Bimbó duality [31] assigns to an OML  $A$  a compact space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), T),$$

where  $F(A)$  is the set of *filters* of  $A$ , ordered by  $\subseteq$  and carrying an orthogonality relation  $\perp_A$  and a Stone topology  $T$ , and  $P(A) = \{\uparrow p : p \in A\} \subseteq F(A)$ , with  $\uparrow p = \{a \in A : a \geq p\}$ , is the set of *principal* filters — the “physical points,” the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For  $a \in A$  set  $h(a) = \{x \in F(A) : a \in x\}$ ; then  $h$  is an OML isomorphism onto the  $\perp$ -stable clopens  $\text{CO}(S_0(A))^\dagger$ . Define  $\mu(h(a)) = s(a)$ . Here  $h(a)$  is the set of points at which  $a$  holds and  $\mu$  is the state transported onto those sets; the open problem is whether  $\mu$  is preserved under countable operations on them.

**Definition 2.13** (Charge). A *charge* on  $S_0(A)$  is a map  $\nu : \text{CO}(S_0(A)) \rightarrow [0, 1]$  on the *full* Boolean algebra of clopen subsets (not merely the  $\perp$ -stable ones in the image of  $h$ ), with

$$\nu(S_0(A)) = 1, \quad \nu(U \cup V) = \nu(U) + \nu(V) \quad \text{whenever } U \cap V = \emptyset.$$

A state  $s$  *extends to a charge* if some such  $\nu$  satisfies  $\nu(h(a)) = s(a)$  for all  $a \in A$ . Because  $h$  preserves meets, a pairwise-meet-zero family in  $A$  maps to disjoint clopens, so a charge necessarily satisfies  $\sum_i s(a_i) \leq 1$  over such families — the charge-extendibility condition of Definition 2.9(3).

*Remark 2.14* (The duality is finitary). The representation map  $h$  satisfies

$$\begin{aligned} h\left(\bigvee_{i \leq n} a_i\right) &= \left(\bigcup_{i \leq n} h(a_i)\right)^{\perp\perp} \quad (\text{finite}), \\ h\left(\bigvee_n a_n\right) &\neq \left(\bigcup_n h(a_n)\right)^{\perp\perp} \quad (\text{countable}) : \end{aligned}$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since  $a \mapsto a^\perp$  is an order anti-isomorphism,  $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$ , so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries  $P(A)$  as explicit structure.

**2.4. The Boolean engine: Stone duality.** In the Boolean case the descent equivalence

$$\sigma\text{-additive} \iff \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [30]:

- (S1) a Boolean event algebra  $B$  embeds in the clopen algebra  $\text{CO}(\text{St } B)$  of its Stone space;
- (S2) a charge on  $B$  extends to a Baire measure on  $\text{St } B$  (Carathéodory);
- (S3)  $\sigma$ -additivity  $\iff$  the measure vanishes on meagre Baire sets (Rao–Rao [37]);
- (S4) meagre-null  $\iff$  the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \cong (\sigma\text{-tribe of sets}) / (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact: the Boolean engine (Stone duality together with the Loomis–Sikorski representation) has *no* known OML analogue (§4.3). Supplying such an engine, or proving none exists, is the open problem of §5.

*Remark 2.15* (Why the bridge breaks: meet-closed but not join-prime). A “realised” point  $x$  carries two conditions, which Rao–Rao’s “vanishes on meagre sets” makes coincide in the Boolean case:

(M) closed under countable meets;

(J)  $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$  for some  $n$  (off the join-defect).

Non-distributivity separates them. For a principal filter  $x = \uparrow p$ , condition (M) always holds, whereas (J) can fail:  $p \leq \bigvee_n a_n$  does not force  $p \leq a_n$  for some  $n$ . In  $L(H)$  this is witnessed by  $a_n = \text{span}(e_n)$  and  $p = \text{span}(e_1 + e_2)$ :

$$p \leq \bigvee_n a_n = H, \quad \text{yet} \quad p \not\leq a_n \text{ for every } n.$$

So  $\uparrow p$  is a realised point lying inside the join-defect  $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$ . Topologically  $\uparrow p$  is *isolated* in  $S_0(L(H))$ , hence:

$$D \text{ non-meagre (OML)} \quad \text{vs.} \quad D \text{ nowhere dense (Boolean)}.$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same  $D$  as the measure route, and reduces to the MacNeille completion of §4.3(ii).

### 3. EXTENSION AND DESCENT

The central question — *when does a state  $s$  on an OML  $A$  extend to a  $\sigma$ -additive measure on  $S_0(A)$ ?* — splits into two *independent* axes that the Boolean case fuses.

*Extension.:*

Does  $s$  extend to a finitely additive charge on the *full* Boolean clopen algebra of  $S_0(A)$ ? This is governed by the meet-zero/orthogonal gap (§2.1) and is *settled*: it is a Horn–Tarski/Pitowsky feasibility question, a decidable linear program in the finite case, and can fail for every state.

*Descent.:*

Once extended, does the charge concentrate on the physical points  $P(A)$ ? This is governed by  $\sigma$ -additivity — it needs the missing engine (§2.4) — and is *open*.

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [13]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [34]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [23]). It is also the marginal problem of probability theory — extending consistent marginals to a joint law (Kellerer [25]) — and the coherence/extendability question of imprecise probability (Walley [43]); the equivalence of these formulations to the quantum-logical one is made precise by Derr–Williamson [9]. What is distinctive here is only that the extension is sought against the dual  $S_0(A)$  rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

**Proposition 3.1** (Finite case). *For finite  $A$  the extension condition is a decidable linear program (Horn–Tarski [23] / Pitowsky correlation-polytope feasibility [34]; Budroni–Morchio [2] for the partial-Boolean-algebra formulation); since  $h$  preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy  $\sum_i s(a_i) \leq 1$  over pairwise-meet-zero families. It can fail for every state (MO<sub>3</sub>, Example 2.10).*

**Proposition 3.2** (Infinite  $L(H)$ , all states). *Let  $\dim H = \infty$ . No state on  $L(H)$  — normal or singular — extends to a charge on  $S_0(L(H))$ .*

*Proof.* The argument embeds a copy of MO<sub>2</sub> in  $L(H)$  whose four atoms are infinite-dimensional, so that the obstruction reaches the singular states as well. Write  $H = H_0 \otimes \mathbb{C}^2$  with  $H_0$  infinite-dimensional, and fix an orthonormal basis  $e, f$  of  $\mathbb{C}^2$ . Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^\perp = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e+f), \quad a_2^\perp = H_0 \otimes \mathbb{C}(e-f).$$

These four infinite-dimensional subspaces form a copy of MO<sub>2</sub> in  $L(H)$ :

$$\begin{aligned} a_1 \perp a_1^\perp, \quad a_2 \perp a_2^\perp, \quad a_i \vee a_i^\perp &= H; \\ a_i \wedge a_j &= \mathbf{0} \quad \text{but} \quad a_i \not\perp a_j \quad (\text{cross pairs}). \end{aligned}$$



Orthoadditivity with  $s(\mathbf{1}) = 1$  gives  $s(a_i) + s(a_i^\perp) = 1$  for  $i = 1, 2$ , using only additivity on orthogonal pairs, so it holds for *every* state. The family  $\{a_1, a_1^\perp, a_2, a_2^\perp\}$  is pairwise meet-zero, and since  $h$  preserves meets ( $h(a \wedge b) = h(a) \cap h(b)$ ), its members map to disjoint clopens; hence any charge satisfies  $\sum_i s(a_i) \leq 1$ . The two counts collide:

$$\sum_i s(a_i) = 2 \quad (\text{orthoadditivity}) \quad \text{versus} \quad \sum_i s(a_i) \leq 1 \quad (\text{any charge}).$$

The construction is finite ( $n = 4$ );  $\sigma$ -completeness of the lattice plays no role.  $\square$

The  $\text{MO}_2$  count is standard (orthoadditivity together with  $s(\mathbf{1}) = 1$ ; see [24]) and  $\sum s(a_i) = 2$  is already state-independent in  $L(\mathbb{C}^2)$ . The role of the *infinite-dimensional* realisation is to extend the obstruction to the singular states ( $s(e) = 0$  on every finite-rank  $e$ ):

| Atoms of $\text{MO}_2$                      | $s(a_i) + s(a_i^\perp)$                   | Reaches                    |
|---------------------------------------------|-------------------------------------------|----------------------------|
| lines in $L(\mathbb{C}^2)$ (finite rank)    | $= 1$ only if $s$ charges finite rank     | normal states only         |
| $H_0 \otimes (\cdot)$ , $\dim H_0 = \infty$ | $= 1$ regardless of finite-rank behaviour | all states, incl. singular |

Keeping each atom infinite-dimensional drops the finite-rank precondition of the line argument (Remark 3.3), and that is what closes the extension axis on  $L(H)$  completely.

*Remark 3.3* (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the  $\text{MO}_2$  family. If  $s(e) > 0$  on some finite-dimensional  $e$ , it is positive on a 2-plane  $e_0$ ; taking  $k$  pairwise-non-orthogonal lines  $p_1, \dots, p_k \subset e_0$  yields  $2k$  pairwise-meet-zero lines with  $s(p_i) + s(p_i^\perp) = s(e_0)$ , so a charge forces

$$k s(e_0) \leq 1 \quad \text{for all } k \quad \implies \quad s(e_0) = 0,$$

contradicting  $s(e_0) > 0$ . Lifting  $s$  to a functional via Mackey–Gleason / Bunce–Wright [3] ( $B(H)$  has no type  $\text{I}_2$  summand) and applying the Takesaki normal/singular decomposition identifies which states the argument reaches: it covers states with a nonzero normal part ( $s(e) > 0$  for some finite-rank  $e$ ), including every normal (Gleason) state, and misses the purely singular states ( $s(e) = 0$  for all finite-rank  $e$ ; e.g. ultrafilter vector states or Calkin-algebra pullbacks). Proposition 3.2 subsumes the missed case; this argument is recorded because its technique is independent and locates the singular states precisely.

#### 4. WHAT IS KNOWN

The extension axis is settled — on  $L(H)$  no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

**4.1. The Boolean benchmark.** The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [14], Thm 391D) is complete: a Boolean algebra carries a strictly positive  $\sigma$ -additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

| Boolean (KVP)<br>condition              | OML analogue                                             |
|-----------------------------------------|----------------------------------------------------------|
| Dedekind $\sigma$ -completeness         | exists ( $\sigma$ -orthocompleteness)                    |
| weak $(\sigma, \infty)$ -distributivity | none — relies on $\bigwedge$ distributing over $\bigvee$ |
| chargeability                           | none — no structural characterisation on OMLs            |

**4.2. Known results for general OMLs. Positive** (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [16] — on  $L(H)$  for a *separable* Hilbert space of  $\dim \geq 3$ , every countably additive measure on the projection lattice has the trace form  $s(\cdot) = \text{tr}(\rho \cdot)$ :  $\sigma$ -additivity *on the lattice*.
- Bunce–Wright [3] — the projection-lattice (JBW) form; Chetcuti–Dvurečenskij [6].

**Obstructive** (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [36] — a *finitary* condition (*subadditive*-unitality  $s(a \vee b) \leq s(a) + s(b)$ , no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the *OMposet* version fails (Müller’s set-representable counterexample [32]).
- Navara [33] — regularity does not force  $\sigma$ -additivity; with Rüttimann’s decomposition of states into completely additive and purely finitely additive parts [38], this shows the finitely-additive/ $\sigma$ -additive gap is structurally serious on quantum logics, not a routine strengthening — part of why the open problem is hard. Pták [35] — exotic state spaces via Greechie pasting.

**4.3. Three routes to a  $\sigma$ -OML engine.** The descent argument needs an OML analogue of the Boolean engine — a  $\sigma$ -Stone duality, equivalently a Loomis–Sikorski representation. Routes (i),(ii) are closed by *theorem*; route (iii) is open in the strong sense (no construction known, none ruled out), and is the Open Problem of §5.

- (i) **RDP / effect-algebra.** Loomis–Sikorski holds for  $\sigma$ -MV algebras [11] and for monotone  $\sigma$ -complete effect algebras *with* the Riesz Decomposition Property [12]; but a lattice effect algebra has RDP iff it is MV, and  $\text{OML} \cap \text{MV} = \text{Boolean}$  (Kalmbach [24]):

OML + RDP  $\iff$  Boolean. The *sharp-skeleton* evasion (represent the OML as the sharp elements of an ambient RDP effect algebra) also closes: in an RDP effect algebra the sharp elements coincide with the center, hence are Boolean (Jenča 2001, Cor. 4.3, via Greechie–Foulis–Pulmannová [17]).

- (ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [20]), so one cannot complete to absorb countable joins; the topological version (clopens of  $S_0(A)$  modulo the meagre ideal) is the same completion and collapses non-distributivity (Remark 2.15).
- (iii)  **$\sigma$ -Stone duality.** None is known, none ruled out: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation (Freytes [15]), but no theorem forbids a countable-join-preserving one. The one machinery supplying a point-free  $\sigma$ -additive valuation, topos/Bohrification, *sidesteps* this route ( $\sigma$ -additivity on a distributive object / per-context Boolean blocks, §4.5) rather than settling it.

**4.4. Prior art on point-free quantum probability.** Every existing point-free or directed-context construction fails at least one of the two requirements —  $\sigma$ -additivity and a natively non-distributive structure — as tabulated below.

**Gleason tradition:**

Varadarajan [41], Kalmbach [24], Pták–Pulmannová [36]:  $\sigma$ -additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

**Döring 2009:**

[10] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

**Bohrification (HLS):**

[21, 22] point-free and the internal valuation *is*  $\sigma$ -additive (Scott-continuous) — but factors through an internally *distributive* locale,  $\sigma$ -additive only per commutative context. Non-distributivity is tamed, not carried natively.

**Localic valuations:**

Vickers [42], Simpson [39], Coquand–Spitters [7]: point-free  $\sigma$ -additive, but on *frames* — distributive by definition.

**OML Stone dualities:**

McDonald–Bimbó [31] and Cannon–Döring [5]: purely structural, no measure, no  $\sigma$ -version.

The open case is their conjunction, met by no existing construction:

point-free  $\times$   $\sigma$ -additive  $\times$  natively non-distributive  $\times$   
directed-system-built.

*Remark 4.1* (The boundary of the negative results). Two qualifications delimit what is ruled out.

- (i): **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$L(H)$  ( $\dim \geq 3$ ): no two-valued states at all (Kochen–Specker [27]),  
a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

- (ii): **No  $\sigma$ -Loomis–Sikorski exists**, and this is visible in the dualities: both take infinite joins as a closure, not a union,

Cannon–Döring:  $\text{cls}(\bigcup S_i)$ ; McDonald–Bimbó:  $(\bigcup h(a_n))^{\perp\perp}$ .

This is not to be overstated: Gudder’s concrete logics [18] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a  $\sigma$ -Loomis–Sikorski theorem.

**4.5. The topos approach.** The topos / Bohrification programme is the one machinery supplying a point-free *and*  $\sigma$ -additive valuation on a quantum logic, so it is the natural candidate for route (iii). In both principal constructions the  $\sigma$ -additive valuation lives on a *distributive* object, its countable additivity routed through per-context Boolean blocks; this is the topos analogue of the universal floor (§5.1).

|                  | Bohrification (HLS)                                                                                                 | Döring–Isham                                                          |
|------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| measure on       | internal Gelfand spectrum $\Sigma(A)$ , a compact regular frame                                                     | clopen subobjects $\text{Sub}_{cl}(\Sigma)$ , a Heyting algebra       |
| why distributive | a frame: $\bigvee$ distributes [22, Def. 6.11]                                                                      | a locale, not a $\sigma$ -algebra [10]                                |
| how Proj enters  | Thm 6.19 bijection; the measure restricts to a $\sigma$ -Boolean morphism on each Boolean sublattice (Def. 6.17(a)) | daseinisation, which “cannot preserve both [joins and meets]” [44]    |
| additivity left  | block-by-block, glued by naturality; never crosses a non-commuting join                                             | only <i>finitely</i> additive; $\sigma$ per-block, normal states only |

In neither case does the additivity cross a non-distributive join; HLS confines all of it inside Boolean contexts, the antithesis of  $\sigma$ -essential contextuality, and its non-distributive object  $\text{Proj}(A)$  is the non-concrete Gleason /  $L(H)$  lattice (Rmk 4.1), not the concrete class in question. The route thus lands on the finitely-additive-versus- $\sigma$  gap exactly: of the four requirements (point-free,  $\sigma$ -additive, natively non-distributive, directed-system-built) it satisfies three

and misses native non-distributivity. It *sidesteps* the wall — an internally distributive Gelfand spectrum is a design goal — rather than settling the  $\sigma$ -essential question, which stays open.

## 5. THE OPEN PROBLEM

*Open Problem.* Does there exist a concrete,  $\sigma$ -complete, non-Boolean OML  $L$  and a  $\sigma$ -additive state  $w$  on  $L$  such that

$$\begin{aligned} (\text{contextual}) \quad & w \notin \overline{\text{conv}}(S_{\text{df}}^\sigma); \\ (\sigma\text{-essential}) \quad & w|_F \in \overline{\text{conv}}(S_{\text{df}}(F)) \text{ for every finite sub-OML } F \subseteq L? \end{aligned}$$

That is,  $w$  has no global hidden joint over the dispersion-free states, yet its restriction to every finite sub-OML does. Equivalently: can one prove no such  $(L, w)$  exists?

*Remark 5.1* (Relation to Derr–Williamson 2023). The problem above is not plainly open: it must be read against Derr–Williamson [9], who study coherent probabilities on *pre-Dynkin-systems* (concrete orthomodular posets) and *Dynkin-systems* (concrete  $\sigma$ -complete OMLs) from the imprecise-probability side. Three distinct levels of statement are involved, and should not be conflated:

- (a) a finite-additive extendability criterion (their Theorem 4.5, a Horn–Tarski feasibility condition) matching “non-contextual on every finite sub-OML”;
- (b) a *countably additive* extension to the ambient Borel  $\sigma$ -algebra (their Theorem D.6), valid *under the hypotheses* that the OML is realised as a Dynkin-system  $D_\sigma$  of Borel sets of a *Polish* space  $\Omega$  with  $\sigma(D_\sigma) = \text{Borel}(\Omega)$  and inner-regular  $\sigma$ -blocks — a result they obtain by specialising Maharam’s extension theorem [29, Thm 8.1] (Hausdorff base, inner-regular measures); and
- (c) the inference from (b) to non-contextuality of  $w$ , below, which is the survey’s own step.

The link to the present formulation is one-directional but in the direction that matters. A point  $\omega \in \Omega$  gives the evaluation  $\delta_\omega$ , which is a two-valued and (since  $A_n \downarrow \emptyset \Rightarrow \omega \notin A_n$  eventually)  $\sigma$ -additive dispersion-free state; a countably additive  $\nu$  on  $\text{Borel}(\Omega)$  with  $\nu|_{D_\sigma} = w$  exhibits  $w$  as a barycentre,

$$w(A) = \int_{\Omega} \delta_\omega(A) d\nu(\omega) \implies w \in \overline{\text{conv}}(S_{\text{df}}^\sigma).$$

Thus

$$\sigma\text{-extendable} \implies \text{non-contextual},$$

and through Theorem D.6 the implication “finitely non-contextual everywhere  $\Rightarrow$  globally non-contextual” holds *whenever the OML admits such a Polish Borel realisation*: under that hypothesis the  $\sigma$ -essential cell is empty, and no witness exists. (The converse, non-contextual  $\Rightarrow$   $\sigma$ -extendable, does not

hold — the dispersion-free states properly contain the point evaluations  $\delta_\omega$  — but it is not needed for the negative conclusion.)

The residue is then a single condition, the representability the whole problem turns on: can a  $\sigma$ -essential witness exist that admits *no* Polish Borel realisation ( $S_{\text{df}}^\sigma$  non-Polish / analytic-but-not-Borel)? This is a descriptive-set-theory question, and it is the same wall as the absence of a  $\sigma$ -Loomis–Sikorski representation (§4.3); the off-centre/non-band constraints of §6 are its lattice-side shadow. Beware the  $\sigma$ -version of the (A)/(B) point-space equivocation (Remark 5.6): Derr–Williamson’s Polish hypothesis is on the underlying *point set*  $\Omega$ , and obtaining such an  $\Omega$  for an abstract concrete  $\sigma$ -complete OML is itself the missing engine, not a free input.

Here “relational” means *no hidden realisation*: the measure is not recoverable from an auxiliary space of definite states behind the propositions. Three point-spaces must be kept distinct: the *canonical* dual  $P(A) = \{\uparrow p\}$  (one principal filter per element, §2.3); the *dispersion-free states* (the hidden-variable points); and the *external (A)-points* (Gleason’s Hilbert rays, Remark 5.6).  $P(A)$  is *permitted*; the objection is only to a measure built on a smuggled space — the (A)-points, equivalently a global hidden-variable joint over the dispersion-free states. By Lemma 5.3 a state has a hidden realisation iff it is a mixture of dispersion-free states, so a relational measure is precisely a *contextual* state. This is a condition on the dispersion-free states, not on  $P(A)$ : a witness is point-rich on  $P(A)$  yet has no global dispersion-free joint.

### 5.1. Reduction to contextuality.

The universal floor. The naive route — realise the lattice as honest sets so that a measure becomes a measure on a space — is closed for every non-Boolean OML.

**Theorem 5.2** (Universal floor). *A faithful set representation of an OML  $L$  — an injective  $h : L \rightarrow \mathcal{P}(\Omega)$  with  $h(a \wedge b) = h(a) \cap h(b)$ ,  $h(a \vee b) = h(a) \cup h(b)$ ,  $h(a^\perp) = \Omega \setminus h(a)$  (a  $\sigma$ -tribe / Loomis–Sikorski representation when  $\sigma$ -joins are preserved) — exists only if  $L$  is Boolean.*

*Proof sketch.* Set operations  $\cap, \cup, ^c$  on  $\mathcal{P}(\Omega)$  distribute, so the image  $h(L)$  is a distributive sublattice;  $h$  injective and lattice-preserving makes  $L$  isomorphic to a distributive ortholattice, i.e. Boolean.  $\square$

Hence no relational measure can arise from a faithful descent to points. What a concrete OML *does* have (Definition 2.7) is an order-determining family of dispersion-free states; the live question is whether every state is a mixture of them.

The reduction. The following equivalence reduces “no hidden realisation” to a spanning condition on the dispersion-free states. Its content is not new — for a Bell scenario it is Fine’s theorem [13] (a joint distribution exists iff a deterministic hidden-variable model does), and the convex-geometric form is Pitowsky’s correlation-polytope picture [34]; the formulation for a

general (concrete) OML, with dispersion-free states as the deterministic vertices, is the natural lift, in the lineage of the sheaf-theoretic account of contextuality [1].

**Lemma 5.3** (Relational = contextual). *Let  $L$  be a concrete OML with dispersion-free state space  $S_{\text{df}}$ , and  $w$  a state on  $L$ . Then  $w$  admits a hidden realisation — a probability measure on a space of global two-valued valuations of  $L$  inducing  $w$  — if and only if  $w \in \overline{\text{conv}}(S_{\text{df}})$ . We call  $w$  contextual when it is not.*

*Proof sketch.* A global two-valued valuation of  $L$  is exactly a dispersion-free state, so a hidden realisation is a probability measure  $\mu$  on  $S_{\text{df}}$  with  $w(a) = \int \omega(a) d\mu(\omega)$  for all  $a$  — i.e.  $w$  is the barycentre of  $\mu$ . The barycentres of probability measures on the (weak-\* compact) set  $S_{\text{df}}$  are exactly  $\overline{\text{conv}}(S_{\text{df}})$ . The handle is *spanning*, not simplicity: non-uniqueness of  $\mu$  is irrelevant; only its *existence* matters.  $\square$

The object sought is therefore a concrete  $\sigma$ -complete OML with a  $\sigma$ -additive state outside the dispersion-free hull.

Why  $\sigma$ -essential. The finite version of this is already inhabited. Wright’s pentagon [45] is a finite, concrete OML — a Greechie loop of five three-atom blocks, hence a lattice — with an order-determining family of dispersion-free states, yet it carries a non-classical  $\{0, \frac{1}{2}\}$ -valued state lying outside their closed convex hull. (The five intertwining atoms form a 5-cycle in which consecutive atoms share a block, so any dispersion-free state, taking at most one of each adjacent pair to 1, sets at most 2 of the 5 to 1; their hull has vertex-sum  $\leq 2$ , whereas the  $\{0, \frac{1}{2}\}$  state — a legitimate state, each block summing to 1 — has vertex-sum  $\frac{5}{2}$ . This is the KCBS inequality [26].) So “concrete OML with a contextual state” is a 1978 fact, and the general spanning statement is false. What Wright does *not* supply is contextuality that requires the countable structure: the pentagon’s is witnessed by a finite sub-OML (itself). The such a state must therefore be  $\sigma$ -essential — contextual in the  $\sigma$ -complete whole but in no finite sub-OML. This is not a device to evade Wright: “every finite piece has a global section, the countable whole has none” is a *compactness failure* of exactly the type by which countable additivity is not first-order axiomatizable over finitely additive coherence (see Remark 5.4).

Status. No witness has been constructed: the product  $\prod_n \text{MO}_2$  fails (its contextual states are only finitely additive — diffuse states on its central  $P(\mathbb{N})$ , which  $\sigma$ -additivity forces to concentrate on points, killing the contextuality, §5.2); Wright is finite; no other construction is known. On the negative side, the cell is now known to be *empty for every OML admitting a Polish Borel point-realisation* (Derr–Williamson, Remark 5.1). What remains genuinely open is therefore the *non-representable* case: a witness, if one exists, must admit no such realisation — which is precisely the absence of a  $\sigma$ -Loomis–Sikorski representation (§4.3), a descriptive-set-theory question.

*Remark 5.4* (The compactness-failure analogy). The  $\sigma$ -essential requirement places the open problem alongside a known Boolean phenomenon. Over a Boolean algebra, countable additivity of a finitely additive charge is not first-order axiomatizable: an ultraproduct of  $\sigma$ -additive charges can be finitely additive but not  $\sigma$ -additive, so every finite (indeed first-order) test is passed while the global property fails [30]. A  $\sigma$ -essential contextual state would be the non-distributive analogue — finite sub-OMLs all classically interpretable, the obstruction carried only by the countable whole.

**5.2. The combinatorial route.** One might seek the object among concrete  $\sigma$ -orthocomplete OMLs, on the grounds that a state's  $\sigma$ -additivity invokes  $\bigvee_n a_n$  only for orthogonal families (Definition 2.5). This route is settled, and does not produce the object. The property that would make  $\sigma$ -additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\star) \quad \exists p, \exists \{a_n\} \text{ infinite orthogonal, } p \wedge a_n = 0 \ (\forall n), \quad p \not\leq a_n \ (\forall n), \quad (2)$$

and  $(\star)$  is satisfied *trivially* by the plain product  $\prod_n \text{MO}_2$ : take  $a_n = a$  in coordinate  $n$  (zero elsewhere), and  $p = \bigvee_n b_n$  where  $b_n = b$  in coordinate  $n$  (zero elsewhere) — a genuine countable *orthogonal* join, so  $p = (b, b, b, \dots)$  exists by  $\sigma$ -orthocompleteness. Then  $p \wedge a_n = 0$  (coordinate  $n$ :  $a \wedge b = 0$  in  $\text{MO}_2$ ; elsewhere  $a_n = 0$ ) and  $p \not\leq a_n$  (coordinate  $n$ :  $b \not\leq a^\perp$ ). Since  $\prod_n \text{MO}_2$  is concrete,  $\sigma$ -complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies  $(\star)$  and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. So the combinatorial axioms ( $\sigma$ -completeness, concreteness,  $(\star)$ ) are *not* the discriminator; the missing ingredient is non-segregation, which is a point-free notion, not a closure property.

*Remark 5.5* (The witness has no points, and the Boolean boundary). The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML. The witness  $\prod_n \text{MO}_2$  shows the failure concretely and at the level of points:  $\text{MO}_2$  admits *no* two-valued *homomorphism* — each of its four separating two-valued *states* ( $a \mapsto 0, b \mapsto 0$ , etc.) fails  $\omega(a \vee b) \leq \omega(a) + \omega(b)$  since  $a \vee b = \mathbf{1}$ , so none preserves  $\vee$  — and  $\prod_n \text{MO}_2$  inherits this, since its diagonal copy  $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$  is a sub- $\text{MO}_2$  on which any homomorphism would restrict to one of  $\text{MO}_2$ 's (none exists). (This is special to  $\text{MO}_2$  and its products — a generic non-Boolean OML such as  $\mathbf{2} \times \text{MO}_2$  *does* carry homomorphisms, e.g. the projection; the universal obstruction is the distributivity argument, not homomorphism-scarcity.) So the witness is concrete (Definition 2.7: order-determining two-valued *states*) and  $\sigma$ -complete yet has no points in the homomorphism sense at all — the point-free target is forced, not engineered. The same arithmetic records the sharp Boolean boundary: an OML is Boolean iff it admits a *unital* set of *subadditive* states — one with, for each nonzero  $a$ , a subadditive state giving



a value 1 (Pták–Pulmannová [36, Thm 1], for OMLs; the orthomodular-poset version fails). A concrete non-Boolean OML therefore cannot have its separating two-valued states form such a unital set: some fail subadditivity (here all four of  $\text{MO}_2$ 's fail  $\omega(a \vee b) \leq \omega(a) + \omega(b)$ ). The forcing property to Boolean is thus subadditivity — a finitary condition — not  $\sigma$ -additivity.

A worked instance fixes the picture. Navara's construction [33] produces an infinite,  $\sigma$ -orthocomplete, rich OML whose inner block is a finite *stateless* Greechie logic — which is exactly what makes Navara's  $\mathcal{L}$  non-concrete. Replacing the stateless block by  $\text{MO}_2$  (write  $\mathcal{L}_2$ ) removes the only source of non-concreteness: the assembly is a product of a sublogic of a Kalmbach horizontal sum (blocks glued only at  $\{0, 1\}$ , *not* an atom-sharing Greechie loop), and pure horizontal sums of concrete logics are concrete, as are sublogics and products. So  $\mathcal{L}_2$  is concrete,  $\sigma$ -orthocomplete, non-Boolean, and carries  $(\star)$  — yet it is not a new object: it is  $\prod_n \text{MO}_2$  with extra scaffolding, segregated in the same way, and it likewise admits no  $\sigma$ -tribe representation. It is the cautionary example showing that exhibiting  $(\star)$  and concreteness together is easy and does *not* reach the open problem.

The art constrains the open ( $\sigma$ -complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third,  $\sigma$ -Stone duality, which is itself open. What is required is not a new idea in the abstract but a representation in that gap: non-distributive enough to evade routes (i),(ii), yet  $\sigma$ -faithful where the known dualities are merely finitary.

*Remark 5.6* (Why  $L(H)$ +Gleason does *not* already settle this — the (A)/(B) point-space equivocation). “Probability from relations, point-free” may *look* delivered by  $L(H)$ +Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on the second. The two are (A) the Hilbert rays of  $H$ , which Gleason *requires* — the density  $\rho$  in  $s = \text{tr}(\rho \cdot)$  is built from them — and (B) the McDonald–Bimbó dual filters  $P(A)$ , which Gleason *removes*, but only by being a representation theorem. Gleason is thus the *most* point-ful result available (every lattice state corresponds to the point datum  $\rho$ ). What  $L(H)$  exhibits is a *separation*, not point-freeness:  $\sigma$ -additive measures exist on the lattice (Gleason, for normal states), yet on the dual  $S_0(L(H))$  no state extends to a charge at all (Proposition 3.2), so a fortiori none concentrates on  $P(A)$ . It cannot serve as the existence proof required, so the point-free  $\sigma$ -additive structure of §5 is the only candidate engine.

**5.3. Two axes of obstruction.** The recurring confusion of this subject (Remarks 5.5, 5.6) is best held as a  $2 \times 2$  table on two *independent* axes, which must not be collapsed into one.

- **Axis 1 — distributivity = commutativity.** One condition in two dialects: a family of projections commutes *iff* the lattice they

generate is distributive. “Commutativity” is the operator word, “distributivity” the lattice word, for the *same* dividing line — not different sides.

- **Axis 2 — concreteness.** A non-Boolean lattice can be *point-rich in rays* (atoms = 1-dim subspaces) yet *point-free in valuations* (no two-valued homomorphisms, Kochen–Specker), or the reverse: the two senses of “point” come apart (the (A)/(B) equivocation, Rmk 5.6).

|                                                     | <b>distributive<br/>(commuting)</b>                              | <b>non-distributive<br/>(non-commuting)</b>                                                                                               |
|-----------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>concrete</b><br>(valuations<br>span)             | classical: Boolean<br>$\sigma$ -algebra,<br>Loomis–Sikorski free | the descent witness lives here: a<br>concrete $\sigma$ -complete<br>non-Boolean OML ( $\prod_n \text{MO}_2$<br>lies here, but segregated) |
| <b>non-<br/>concrete</b><br>(KS: few<br>valuations) | atomless measure<br>algebra (valuation-poor)                     | $L(H)$ : rays abundant,<br>valuations KS-barred — the<br>operator / e.s.a. world, the<br>excluded pole                                    |

There is not one “lattice  $\leftrightarrow$  space” dictionary but two, one per row: *Birkhoff–von Neumann* (bottom row) sends elements to closed subspaces and atoms to **rays**, with  $a \vee b$  the closed span — rich in ray-points, KS-barred from valuation-points, hence non-concrete; *Stone / concrete-logic* (Gudder, top row) sends elements to sets of outcomes and atoms to points of  $\Omega$  — rich in valuation-points, hence concrete. The operator question and the descent question share Axis 1 but sit in opposite rows: spectral projections *force*  $L(H)$  (bottom-right), whereas the descent witness must be top-right. So the descent problem is the one cell no operator algebra can occupy — why a witness must be a pasting / countable colimit of concrete blocks, not a Hilbert-space construction, and why  $L(H)$ +Gleason cannot settle it. In this language the open problem (route (iii)) is exactly: *is there a logic  $\leftrightarrow$  geometry dictionary for the top-right cell?* — and a  $\sigma$ -Loomis–Sikorski for concrete OMLs is that missing dictionary. Axes 1 and 2 are independent: non-distributivity does not entail non-concreteness, and a witness must be concrete and non-distributive simultaneously.

## 6. DIRECTIONS

With the extension axis closed on  $L(H)$  (Proposition 3.2), the combinatorial route shown to deliver only segregated objects (§5.2), and the question reduced to a  $\sigma$ -essential contextual state (§5.1), the problem has a clean dichotomy: two exits, each a genuine terminus. The negative exit is, moreover, *already taken in the Polish-representable case* (Derr–Williamson, Remark 5.1); both exits below therefore live or die on the same feature — whether a witness can fail to admit a regular Borel point-realisation.

**Exhibit a  $\sigma$ -essential witness:**

Construct an infinite concrete OML carrying a  $\sigma$ -additive contextual state that no finite sub-OML witnesses. The natural attack is a compactness-failure construction (cf. Remark 5.4): a concrete  $\sigma$ -complete OML whose finite sublogics are all classically interpretable (dispersion-free states span them) but whose countable joins force a Kochen–Specker-type obstruction only in the limit. The faithful set representation is ruled out at every stage (Theorem 5.2), so the obstruction must be carried by the  $\sigma$ -structure itself, not by any finite block. A necessary condition follows: the witness must place its infinitary structure *off-center* (irreducible, with non-central infinite blocks). A product of finite blocks such as  $\prod_n \text{MO}_2$  is excluded a priori, since it forces all infinitary content into the Boolean center, where  $\sigma$ -additivity concentrates the contextual states onto points and removes them (§5.2); a pasting or countable colimit of Wright-type blocks is the natural candidate.

**Prove no such witness exists:**

A spanning theorem: every  $\sigma$ -additive state on a  $\sigma$ -complete concrete OML lies in the closed convex hull of its dispersion-free states *except* where a finite sub-OML is already contextual (Wright). This would be an impossibility result — a  $\sigma$ -essential contextual state cannot exist, so relational  $\sigma$ -additive probability is operationally invisible (present finitely or not at all). *This is already established for Polish-representable OMLs* (Derr–Williamson, Remark 5.1); a full impossibility result must extend it to the non-representable case, where the Polish/Borel-regularity hypothesis of their Theorem D.6 is unavailable. The Boolean boundary — subadditivity of a unital set of states (Pták–Pulmannová, Remark 5.5) — is a natural lever; note that such an argument cannot reuse the  $\prod_n \text{MO}_2$  mechanism (concentration on the Boolean center), which is vacuous off-center.

A candidate direction now closed: the singular case for  $L(H)$ . One might have hoped a singular state would extend where no normal state does, providing the first explicit “extends but may fail to concentrate” example; Proposition 3.2 rules this out, catching every state, singular included. The reduction is recorded because the singular case was expected to require infinitary methods and does not (Remark 6.1).

*Remark 6.1* (Finite versus countable: finite suffices). Is the obstruction to extending a singular state realised by a *finite* or only by a *countable* pairwise-meet-zero family? The two arguments divide on exactly this point. The line argument (Remark 3.3) is finitary but *vacuous* on a singular state ( $s(e) = 0$

for every finite-rank  $e$ ); the  $\text{MO}_2$  argument with infinite-dimensional atoms (Proposition 3.2) uses a finite family ( $n = 4$ ) and gives  $\sum_i s(a_i) = 2$  state-independently, catching the singular case as well. So the singular obstruction is *not* entangled with the descent-axis passage, as the vacuity of the line argument might suggest: it is realised by a finite family, hence a self-contained finitary problem, separable from the general construction and not governed by the finitary-to- $\sigma$  passage.

## APPENDIX A. GLOSSARY OF DEFINITIONS

A self-contained list of the notions used in the paper. Entries that re-state a numbered definition in the body cross-reference it; the remaining entries are stated here. Throughout,  $L$  is an orthomodular lattice with orthocomplementation  $a \mapsto a^\perp$ , bottom  $\mathbf{0}$ , top  $\mathbf{1}$ .

### Lattice notions.

#### Orthocomplemented lattice; orthomodular lattice (OML).:

A bounded lattice with an order-reversing, involutive complement  $a^\perp$  satisfying  $a \vee a^\perp = \mathbf{1}$ ,  $a \wedge a^\perp = \mathbf{0}$  (Definition 2.1); it is *orthomodular* if  $a \leq b \Rightarrow b = a \vee (b \wedge a^\perp)$  (Definition 2.4). A *Boolean algebra* is a distributive OML.

#### Orthogonal; meet-zero; incomparable.:

$a \perp b$  means  $a \leq b^\perp$ ; *meet-zero* means  $a \wedge b = \mathbf{0}$ . Orthogonal implies meet-zero; the converse fails off the Boolean case, and that gap drives the subject (Definition 2.2).

#### Compatible; block.:

$a, b$  are *compatible* if they generate a Boolean subalgebra, equivalently  $a = (a \wedge b) \vee (a \wedge b^\perp)$ . A *block* is a maximal pairwise-compatible subset of  $L$ ; it is a maximal Boolean sub-OML, and every OML is the union of its blocks. (In a concrete logic, compatibility of  $A, B$  is  $A = (A \cap B) \cup (A \cap B^c)$ .)

#### Completeness.:

$L$  is  $\sigma$ -*complete* if every countable subset has a join; it is  $\sigma$ -*orthocomplete* if every countable *pairwise-orthogonal* family has a join (Definition 2.5).

#### Centre; irreducible.:

The *centre*  $Z(L)$  is the set of elements compatible with all of  $L$ ; it is a Boolean subalgebra.  $L$  is *irreducible* if  $Z(L) = \{\mathbf{0}, \mathbf{1}\}$  (no nontrivial direct-product decomposition).

#### $\text{MO}_n$ ; $L(H)$ .:

$\text{MO}_n$  is the modular ortholattice with  $\mathbf{0}, \mathbf{1}$  and  $n$  complementary atom-pairs ( $\text{MO}_2$ : atoms  $a, a^\perp, b, b^\perp$ , all distinct meets  $\mathbf{0}$ ); it is non-distributive for  $n \geq 2$ .  $L(H)$  is the lattice of closed subspaces of a Hilbert space  $H$  (orthocomplement = orthogonal complement, join = closed span), the prototype non-distributive OML.

## Representations.

### OMP; concrete logic; order-determining states.:

An *orthomodular poset* (OMP) weakens an OML to a partial join (only of orthogonal elements).  $L$  is a *concrete logic* (set-representable) if it embeds in some  $\mathcal{P}(\Omega)$  with  $a^\perp \mapsto \Omega \setminus a$  and orthogonal joins as disjoint unions, equivalently if it carries an *order-determining* (separating) family of two-valued states (Definition 2.7).

### Faithful set representation; $\sigma$ -tribe.:

An injection  $h : L \rightarrow \mathcal{P}(\Omega)$  with  $h(a \wedge b) = h(a) \cap h(b)$ ,  $h(a \vee b) = h(a) \cup h(b)$ ,  $h(a^\perp) = \Omega \setminus h(a)$  (preserving *all* meets and joins as  $\cap, \cup$ ); a  $\sigma$ -tribe / Loomis–Sikorski representation also preserves countable joins. By Theorem 5.2 only Boolean  $L$  admits one. (Contrast a concrete logic, which preserves only *orthogonal* joins as unions.)

### Dual space $S_0(A)$ ; canonical points $P(A)$ .:

$S_0(A)$  is the McDonald–Bimbó space of filters of  $A$ ;  $P(A) = \{\uparrow p : p \in A\}$  is the set of principal filters, one per element (Definition 2.12).

## States and measures.

### State; meet-additive; charge.:

A *state* is  $s : L \rightarrow [0, 1]$  with  $s(\mathbf{1}) = 1$  and  $s(a \vee b) = s(a) + s(b)$  for  $a \perp b$  (Definition 2.9); a *charge* is a finitely additive set function on a Boolean algebra (Definition 2.13).

### $\sigma$ -additive state.:

A state  $s$  with  $s(\bigvee_n a_n) = \sum_n s(a_n)$  for every countable pairwise-orthogonal  $\{a_n\}$  (Definition 2.5).

### Dispersion-free state.:

A two-valued state, i.e. a state taking only the values 0, 1 (equivalently a  $\{0, 1\}$ -homomorphism on each block);  $S_{\text{df}}$  denotes their set and  $S_{\text{df}}^\sigma$  the  $\sigma$ -additive ones. These are the deterministic / hidden-variable points.

### Normal; singular state on $L(H)$ .:

$s$  is *normal* if  $s(\cdot) = \text{tr}(\rho \cdot)$  for a density operator  $\rho$ ; *singular* if  $s$  vanishes on every finite-rank projection (Definition 2.11).

### Subadditive state.:

A state  $s$  with  $s(a \vee b) \leq s(a) + s(b)$  for *all*  $a, b$  (not only orthogonal pairs). A family of states is *unital* if for each  $a \neq \mathbf{0}$  some member gives  $s(a) = 1$ .

## The descent vocabulary.

### Contextual state.:

A state  $w$  that is *not* in the closed convex hull  $\overline{\text{conv}}(S_{\text{df}})$  of the dispersion-free states — equivalently (Lemma 5.3) a state with no hidden realisation, i.e. not a mixture of dispersion-free states. A non-contextual state is one that is such a mixture.

**$\sigma$ -essentially contextual state.:**

A  $\sigma$ -additive state  $w$  on a  $\sigma$ -complete  $L$  that is *contextual* —  $w \notin \overline{\text{conv}}(S_{\text{df}}^\sigma)$  — yet whose restriction to *every finite sub-OML* of  $L$  is non-contextual. (All three clauses are required:  $\sigma$ -additive, globally contextual, finitely non-contextual. The finite version without the third clause is already inhabited, Wright’s pentagon.)

**Extension axis; descent axis.:**

The *extension* axis asks whether a state extends to a charge on the full clopen algebra of  $S_0(A)$  (a Horn–Tarski feasibility question); the *descent* axis asks whether such a charge concentrates on the physical points  $P(A)$  (governed by  $\sigma$ -additivity). See §3.

**Segregated.:**

Of an OML: one whose non-distributivity is confined to finite blocks, so that  $S_{\text{df}} = S_{\text{df}}^\sigma$  and no  $\sigma$ -essential leak is possible (e.g.  $\prod_n \text{MO}_2$ , §5.2).

**Prior-art objects (§5, Remark 5.1).****Pre-Dynkin-system; Dynkin-system.:**

In Derr–Williamson’s terminology, a *pre-Dynkin-system* on  $\Omega$  is a concrete OMP (a family of subsets containing  $\Omega$ , closed under complement and under disjoint unions); a *Dynkin-system* additionally is closed under countable disjoint unions — a concrete  $\sigma$ -complete OML.

 **$\sigma$ -extendable.:**

In Derr–Williamson, a countably additive probability  $\mu_\sigma$  on a Dynkin-system  $D_\sigma \subseteq \text{Borel}(\Omega)$  ( $\Omega$  Polish) is  *$\sigma$ -extendable* if it extends to a countably additive probability on  $\text{Borel}(\Omega)$ .

**Polish space; inner regular.:**

A *Polish* space is a separable completely metrizable space. A measure  $\mu$  is *inner regular* if  $\mu(F) = \sup\{\mu(K) : K \subseteq F, K \text{ compact}\}$ .

## REFERENCES

- [1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.
- [2] Costantino Budroni and Giovanni Morchio. The extension problem for partial Boolean structures in quantum mechanics. *Journal of Mathematical Physics*, 51(12):122205, 2010. arXiv:1010.4662.
- [3] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.
- [4] Dominika Burešová and Pavel Pták. On the set-representable orthomodular posets that are point-distinguishing. *International Journal of Theoretical Physics*, 62, 2023. Also arXiv:2401.13798.
- [5] Sarah Cannon. *The Spectral Presheaf of an Orthomodular Lattice: A Generalisation of Stone Duality to Orthomodular Lattices*. PhD thesis, University of Oxford, 2013. See also Cannon and Döring, “A generalisation of Stone duality to orthomodular lattices,” in *Contemporary Logic and Computing* (2018).
- [6] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.

- [7] Thierry Coquand and Bas Spitters. Integrals and valuations. *Journal of Logic & Analysis*, 1(3):1–22, 2009.
- [8] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.
- [9] Rabanus Derr and Robert C. Williamson. Systems of precision: Coherent probabilities on pre-dynkin-systems and coherent previsions on linear subspaces. *arXiv preprint arXiv:2302.03522*, 2023.
- [10] Andreas Döring. Quantum states and measures on the spectral presheaf. *Advanced Science Letters*, 2(3):291–301, 2009. arXiv:0809.4847.
- [11] Anatolij Dvurečenskij. Loomis–sikorski theorem for  $\sigma$ -complete MV-algebras and  $\ell$ -groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.
- [12] Anatolij Dvurečenskij. Loomis–sikorski representation of monotone  $\sigma$ -complete effect algebras. *Fuzzy Sets and Systems*, 157(5):683–690, 2006.
- [13] Arthur Fine. Hidden variables, joint probability, and the Bell inequalities. *Physical Review Letters*, 48(5):291–295, 1982.
- [14] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras*. Torres Fremlin, Colchester, 2004.
- [15] Hector Freytes. An equational theory for  $\sigma$ -complete orthomodular lattices. *Soft Computing*, 24(14):10257–10264, 2020.
- [16] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.
- [17] Richard J. Greechie, David J. Foulis, and Sylvia Pulmannová. The center of an effect algebra. *Order*, 12(1):91–106, 1995.
- [18] Stanley P. Gudder. *Stochastic Methods in Quantum Mechanics*. North-Holland, New York, 1979.
- [19] Yukio-Pegio Gunji, Taichi Haruna, and Daisuke Uragami. Contextuality as a left adjoint: a categorical generation of orthomodular structure. *arXiv preprint arXiv:2603.22353*, 2026.
- [20] John Harding. Orthomodular lattices whose MacNeille completions are not orthomodular. *Order*, 8(1):93–103, 1991.
- [21] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. A topos for algebraic quantum theory. *Communications in Mathematical Physics*, 291(1):63–110, 2009. arXiv:0709.4364.
- [22] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. Bohrification. In Hans Halvorson, editor, *Deep Beauty: Understanding the Quantum World through Mathematical Innovation*, pages 271–313. Cambridge University Press, 2011. arXiv:0909.3468.
- [23] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.
- [24] Gudrun Kalmbach. *Orthomodular Lattices*. Academic Press, 1983.
- [25] Hans G. Kellerer. Verteilungsfunktionen mit gegebenen marginalverteilungen. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 3:247–270, 1964.
- [26] Alexander A. Klyachko, M. Ali Can, Sinem Binicioğlu, and Alexander S. Shumovsky. Simple test for hidden variables in spin-1 systems. *Physical Review Letters*, 101(2):020403, 2008.
- [27] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.
- [28] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.
- [29] Dorothy Maharam. Consistent extensions of linear functionals and of probability measures. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Vol. II: Probability Theory*, pages 127–147. University of California Press, Berkeley, 1972.
- [30] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.

- [31] Joseph McDonald and Katalin Bimbó. A Stone-type duality for orthomodular lattices. *Mathematical Logic Quarterly*, 2023. arXiv:2208.07430.
- [32] V. Müller. Jauch–piron states on concrete quantum logics. *International Journal of Theoretical Physics*, 32(3):433–442, 1993.
- [33] Mirko Navara. Regularity and  $\sigma$ -additivity of states on quantum logics. *Proceedings of the American Mathematical Society*, 115(2):427–429, 1992.
- [34] Itamar Pitowsky. *Quantum Probability — Quantum Logic*, volume 321 of *Lecture Notes in Physics*. Springer, Berlin, 1989.
- [35] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.
- [36] Pavel Pták and Sylvia Pulmannová. A measure-theoretic characterization of Boolean algebras among orthomodular lattices. *Commentationes Mathematicae Universitatis Carolinae*, 35(1):205–208, 1994.
- [37] K. P. S. Bhaskara Rao and M. Bhaskara Rao. *Theory of Charges: A Study of Finitely Additive Measures*. Academic Press, New York, 1983.
- [38] Gottfried T. Rüttimann. Weakly purely finitely additive measures. *Canadian Journal of Mathematics*, 46(4):872–885, 1994.
- [39] Alex Simpson. Measure, randomness and sublocales. *Annals of Pure and Applied Logic*, 163(11):1642–1659, 2012.
- [40] Josef Tkadlec. Boolean orthoposets — concreteness and orthocompleteness. *Mathematica Bohemica*, 119(2):123–128, 1994.
- [41] Veeravalli S. Varadarajan. *Geometry of Quantum Theory*. Springer, New York, 2nd edition, 1985.
- [42] Steven Vickers. A localic theory of lower and upper integrals. *Mathematical Logic Quarterly*, 54(1):109–123, 2008.
- [43] Peter Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman and Hall, London, 1991.
- [44] Sander Wolters. A comparison of two topos-theoretic approaches to quantum theory. *Communications in Mathematical Physics*, 317(1):3–53, 2013. arXiv:1010.2031.
- [45] Ron Wright. The state of the pentagon: a nonclassical example. In A. R. Marlow, editor, *Mathematical Foundations of Quantum Theory*, pages 255–274. Academic Press, New York, 1978.